

데이터 기반 신소재 공학(공) (MSE308-01)

학기	2026년 2학기
담당교수	박인철
이수구분	교선�필수
학점	3.0
강의시간	월13:00-14:30(E7 - 229), 수13:00-14:30(E7 - 229)
강의실	E6 215 (조별토론실)
소속	기초학부

교과개요

This course examines how data on materials composition, processing, structure, and properties can inform engineering decisions and well-justified next experiments. Students learn to define research questions and measurement criteria, then apply foundational principles of experimental design to generate data that support fair comparisons. The course also covers data-quality assessment, baseline models, appropriate validation, uncertainty, and the limits of interpreting materials data that have already accumulated. Statistical methods and machine learning are used to judge what the evidence supports and to decide what to do next in materials development. They are not treated as a competition for the highest model score. Students work with materials-development cases and instructional datasets through calculation, data analysis, team discussion, and independent verification. By the end of the semester, they should be able to propose a next experiment and a proceed, hold, or stop criterion while stating the limitations of the available data and models. Basic probability and statistics, along with elementary tabular calculations in Python or a spreadsheet, are recommended prerequisites.

교과목표

By the end of the course, students will be able to:

Turn a materials-development question into clear measurements and a clear definition of what one independent data point represents.

Use randomization, replication, blocking, and factorial design to make fair comparisons.

Check where existing data came from, what is missing, and what conclusions the data can support.

Compare simple statistical and machine-learning models using fair tests.

Use uncertainty, physical checks, and reproducibility to judge whether a model result is useful.

Use tables, images, and spectra as evidence, then recommend the next experiment with a clear proceed, hold, or stop criterion.

주차별 강의계획

1주차:How data guide materials development

2주차:Asking a research question and deciding what to measure

3주차:Making fair comparisons: randomization, replication, and blocking

4주차:Seeing how two factors work together

5주차:Checking a result with a follow-up experiment

6주차:Understanding where existing data come from and their limits

7주차:Starting with simple models and inspecting their errors

8주차:Midterm examination

9주차:Testing models fairly and avoiding information leakage

10주차:Comparing ways to represent materials data and choosing a model
11주차:Using uncertainty to make a decision
12주차:Reading materials images at the specimen level
13주차:Using spectra and checking whether model explanations are stable
14주차:Planning the next experiment under real constraints
15주차:Reserve week; individual assignment submission
16주차:Final examination

수업방법

The course combines lectures and presentations with individual calculations, team discussion, data analysis, and project work. Students use materials-development cases and instructional datasets to make decisions, explain the evidence behind them, and independently check their reasoning.

학습윤리

Students must submit their own work and accurately represent data sources, calculations, data processing, and AI use. AI tools, internet searches, and other students' answers are prohibited in examinations, designated independent hand calculations, and individual judgments made before results are revealed. When AI use is permitted in exploratory activities and assignments, students must record the tool, its purpose, the output they used or rejected, and an independent verification step. Submitting unverified AI output as one's own judgment, fabricating data, calculations, sources, or activity records, or copying another student's work is academic misconduct. Examination violations are handled under applicable university procedures.

교과정책(방침)

The course normally meets in person. Students are expected to participate in lectures, individual activities, team discussions, calculations, and data analysis. Teams of three are recommended; teams of two to four may be used depending on enrollment. Assignment formats and exact deadlines will be announced through the course portal. Approved absences, accessibility accommodations, and alternative participation or assessment arrangements follow university procedures. At least one laptop per team and access to Python or spreadsheet software are recommended.

준비물 및 기타

Laptop and access to Python or spreadsheet software are recommended. A PDF of the required course text, Materials Development Methodology: Designed Experiments, Inherited Data, and Next Experiments, will be provided during class. It is not yet a formal edition.